

JS Asssignment

// 1. Write a program using var inside a block and print it outside the block. Explain the output.

```
function check() {  
  if (true) {  
    // Declared inside an 'if' block  
    var message = "block";  
  }  
  // Printed outside the 'if' block  
  console.log("term", message);  
}
```

```
check();
```

//2. Create an object named student with properties name, age, and a function display(). Print all details and call the function.

```
let student = {  
  name: "barani",  
  age: "20",  
  
  display(){  
    console.log("Name is " + this.name + " and age is " + this.age);  
  }  
}
```

```
};
```

```
console.log(student.name);
```

```
console.log(student.age);
```

```
student.display();
```

// 3. Create an array [10, 20, 30, 40, 50] and use map(), filter(), and find() to perform different operations.

```
let arr = [10 , 20 , 30 , 40 , 50];
```

```
console.log(arr.map(map=>map*2));
```

```
console.log(arr.filter(filter=>filter>20));
```

```
console.log(arr.find(find=>find==30));
```

// 4. Write a function that accepts two numbers as parameters and returns their sum, difference, and multiplication.

```
function math(a,b){
```

```
    let sum = a+b;
```

```
    let difference = a-b;
```

```
    let multiplication = a*b;
```

```
    return{
```

```
        sum: sum,
```

```
        difference: difference,
```

```
        multiplication: multiplication
```

```
    };
```

```
}
```

```
let result = math(10,5);
```

```
console.log("Sum:", result.sum);
```

```
console.log("Difference:", result.difference);
```

```
console.log("Multiplication:", result.multiplication);
```

// 5. Create a function that returns another function. Store the returned function in a variable and invoke it.

```
function outerFunction() {
```

```
    let message = "Hello this a inner function";
```

```
    return function innerFunction() {
```

```
        console.log(message);
```

```
    };
```

```
}
```

```
const mySubFunction = outerFunction();
```

```
mySubFunction();
```
